

AI Adoption Analytics Project Documentation

Executive Summary

This project analyzes the evolution of AI adoption among developers worldwide between 2023 and 2025 using the Stack Overflow Developer Survey datasets. The analysis focuses on understanding how developers use AI tools, how they perceive and trust AI technologies, and how AI adoption evolved over time across different demographic groups. The project combines data cleaning, exploratory analysis, KPI development, and dashboard storytelling using Python and Power BI.

Data Sources

Stack Overflow Developer Survey

Official datasets:

- 2023 Developer Survey
- 2024 Developer Survey
- 2025 Developer Survey

Official survey platform:

- [Stack Overflow Developer Survey](#)

Dataset Links

2023 Survey Dataset

- [Stack Overflow Survey 2023 Results](#)

2024 Survey Dataset

- [Stack Overflow Survey 2024 Results](#)

2025 Survey Dataset

- [Stack Overflow Survey 2025 Results](#)

Survey Documentation & Methodology

2023 Survey Documentation

- [Stack Overflow Survey 2023 Methodology](#)

2024 Survey Documentation

- [Stack Overflow Survey 2024 Methodology](#)

2025 Survey Documentation

- [Stack Overflow Survey 2025 Methodology](#)

These datasets contain responses from tens of thousands of developers worldwide covering:

- AI usage
- AI sentiment
- AI trust
- Developer demographics
- AI tools and AI agents

Methodology

1. Data Cleaning

- Handling missing values
- Replacing NaN values with analytical labels
- Standardizing categories across years
- Creating grouped variables for analytics

2. Exploratory Data Analysis

- AI adoption distribution
- Sentiment analysis
- Trust analysis
- Country and demographic analysis
- AI capability analysis
- AI agent adoption analysis

3. Dashboarding

Interactive Power BI dashboards were created to visualize:

- Adoption trends
- Trust evolution
- Sentiment changes
- Demographic comparisons
- Geographic insights

Dashboard Pages

Dashboard 1 — 2023 Baseline Analysis

Focus:

- Early AI adoption
- Initial developer perception
- Baseline trust levels

Dashboard 2 — 2024 Growth Analysis

Focus:

- AI maturity
- Increased adoption
- Perceived AI capabilities

Dashboard 3 — 2025 Advanced AI Era

Focus:

- AI agents
- Advanced AI workflows
- Complex task handling

Dashboard 4 — Evolution Analysis

Focus:

- AI adoption trends (2023–2025)
- Trust evolution
- Sentiment evolution
- Cross-year comparison

Tools & Technologies

- Python
- Pandas
- Jupyter Notebook
- SQL Analytics Logic
- Power BI

Project Outcomes

The project demonstrates:

- Data cleaning and preprocessing
- Exploratory data analysis
- KPI design and dashboard storytelling
- Business insight extraction
- Data visualization best practices

The final result is a professional business intelligence project analyzing one of the most important technological shifts in modern software development: the rise of AI adoption.